

Analyzing Bias Propagation in Recommender Feedback Loops Using Embeddings

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Outline

- 1 Domain of Interest
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Feedback Loop Bias Example

Netflix

Watch Stranger Things



Recommend more Sci-Fi



User watches Sci-Fi



Model learns: "Likes Sci-Fi"



Future recommendations mostly Sci-Fi

Feedback Loop: Recommendations shape user behavior, which is used to retrain the model, reinforcing the same recommendations and reducing diversity.

- **Recommender Systems (RSs):** Digital platforms use recommendation systems to discover user preferences and match users to catalog items.
- **Vector Embeddings:** Recommendation engines map users and items into a shared, low-dimensional space as dense vectors.
- **Closed Feedback Loops:** Recommendation outputs shape future user interactions, which become inputs for future model training.
- **Embedding Geometry:** The study focuses on how vector coordinates move and change over repeated training cycles.

Problem Statement: Feedback Loops and Bias Propagation

- Observational training data has systemic bias and deviates from true user preferences.
- Standard models favor popular items and ignore most items in the catalog.
- Feedback loop bias is dynamic, self-reinforcing, and grows stronger across retraining cycles.
- Users with niche preferences do not receive recommendations matching their real interests.

Problem Statement: Embedding Space Drift

- Embeddings store item attributes and user preference patterns as spatial coordinates.
- Feedback loops cause vector representations to undergo continuous geometric drift.
- Vectors move away from true user preference coordinates toward a biased focal center.
- This geometric drift contracts the embedding space, reducing catalog coverage and recommendation diversity.

Project Objectives

Primary Objective

Analyze and mitigate dynamic bias propagation in closed-loop recommender systems by tracking geometric changes in vector embedding space.

- 1 **Build Simulation Framework:** Construct a closed-loop simulation environment to model multi-generation recommendation cycles.
- 2 **Quantify Vector Contraction:** Calculate the velocity of embedding space contraction to measure diversity loss and popularity concentration.
- 3 **Evaluate Mitigation Methods:** Test alternative recommendation algorithms and bias mitigation techniques to stabilize vector trajectories.

Expected Outcome

- Develop a recommendation model for feedback loop analysis.
- Train and evaluate the model through iterative recommendation cycles.
- Analyze the evolution of embedding representations over time.
- Study the impact of feedback loops on bias and recommendation diversity.
- Provide insights for future bias-aware recommender systems.

Literature Review

Paper	Methodology	Key Findings	Limitations
Zoralioglu & Yalcin (2026)	10-iteration loop framework on benchmarks.	Loops degrade calibration and long-tail item exposure.	Evaluates lists directly; lacks vector-space metrics.
Liu et al. (2026)	CDSRec model using Markov chains, DAGs, and VAEs.	Separates intrinsic quality from trend bias.	High complexity; no coordinate contraction tracking.
Chen et al. (2023)	Survey of debiasing (IPS, Causal, RL).	Identifies feedback loops as primary bias driver.	Qualitative; lacks empirical vector tracking.
Mansoury et al. (2020)	Offline simulation across UserKNN, BPR, Most-Popular.	Loops amplify popularity bias and drop diversity.	Fixed acceptance rules; no latent vector analysis.
Chaney et al. (2018)	Multi-generation agent-based simulations.	Homogenizes users and harms niche preferences.	Relies on synthetic utility; ignores vector drift.

- **Research Gap 1: Static vs Dynamic Bias** – Existing mitigation strategies treat bias as a static problem, ignoring how feedback loops amplify bias over time.
- **Research Gap 2: Structural Vector Analysis** – Existing studies do not measure the geometric drift of vector embeddings during retraining loops.
- **Bridging the Gap:**
 - ▶ Define mathematical metrics to track embedding displacement and contraction speed.
 - ▶ Identify algorithmic choices that prevent latent space collapse.
 - ▶ Preserve catalog coverage and improve recommendations for users with niche preferences.

Thank You!